

# Scientific Diagram Alternative Text Guide

## Writing alt text for molecular structures, reaction schemes, and laboratory diagrams

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#### The problem with generic guidance

Most published advice on alternative text was written for photographs and simple charts. Applied to a scientific diagram it produces something like:

alt="Diagram of a benzene ring"

That is accurate and useless. It tells a student that a diagram exists. It does not tell them what the diagram shows, and the diagram is the content.

A structural formula carries its meaning in **composition, connectivity, and geometry** — which atoms are present, what is bonded to what, and how it is arranged in space. Alternative text that omits those omits the chemistry.

#### The three-part test

Before writing alt text, decide which of three categories the image falls into.

**Decorative.** The image adds visual interest but no information. A stock photograph of a laboratory at the top of a syllabus. Mark it decorative — in Word, tick *Mark as decorative*; in HTML, alt="". Do not describe it.

**Informative.** The image conveys information a student needs, and that information can be stated in a sentence or two. Most single molecules, simple apparatus, and straightforward graphs. Write concise alt text.

**Complex.** The information requires more than about two sentences — reaction mechanisms, multi-step syntheses, phase diagrams, annotated apparatus. Use the **short alt plus long description** pattern described below.

## The short alt plus long description pattern

For complex images, alternative text alone is the wrong tool. Screen readers do not pause well inside long alt strings, and the student cannot re-read part of it.

Instead:

1. Write **short alt text** naming what the image is: alt="Reaction mechanism for the acid-catalysed hydration of ethene, described in the text below."
2. Place the **full description in the body of the page**, immediately after the image, as ordinary readable text.

The description then benefits every student, not only screen reader users. A student who finds the diagram hard to parse can read the prose version. This is the single most useful habit in this guide.

Do not hide the description behind a “click to expand” control unless the control is keyboard accessible and clearly labelled.

## Templates by diagram type

### Single molecule, structural formula

State molecular formula, then the skeleton, then substituents and their positions, then stereochemistry if it matters.

Structural formula of ethanol, C<sub>2</sub>H<sub>5</sub>OH. A two-carbon chain. Carbon one carries three hydrogens. Carbon two carries two hydrogens and a hydroxyl group.

For a ring system:

Structural formula of para-aminophenol. A benzene ring with a hydroxyl group at position one and an amino group at position four, directly opposite.

### Reaction scheme

State reactants, arrow type, conditions, products. Read left to right as drawn.

Reaction scheme. Ethene plus water, over an acid catalyst, yields ethanol. A single forward arrow labelled “H<sub>2</sub>SO<sub>4</sub>, heat.”

For equilibrium, name the arrow explicitly:

Nitrogen gas plus three moles of hydrogen gas is in equilibrium with two moles of ammonia gas. Double half-arrows indicate a reversible reaction. Conditions above the arrows read “iron catalyst, 450 degrees Celsius, 200 atmospheres.”

### Reaction mechanism with curly arrows

Curly arrows show electron movement and are the entire point of the diagram. Describe each arrow as an origin and a destination.

Step one of two. A curly arrow originates at the pi bond between carbon one and carbon two and terminates at the hydrogen of the acid, showing the pi electrons attacking the proton. This forms a carbocation at carbon two, shown with a positive charge.

### Apparatus and laboratory setup

Describe spatially, in the order a student would assemble it.

Distillation apparatus. From left to right: a round-bottom flask on a heating mantle, connected by a still head to a thermometer inserted from above. The still head connects to a water-cooled condenser angled downward, with water entering at the lower inlet and leaving at the upper outlet. The condenser terminates in a receiving flask.

### Graphs and titration curves

State axes with units, overall shape, and the features a student is meant to read off.

Titration curve. The horizontal axis shows volume of sodium hydroxide added, zero to fifty millilitres. The vertical axis shows pH, zero to fourteen. The curve begins near pH 1, rises gradually to about pH 4 by twenty millilitres, then rises steeply through the equivalence point at approximately twenty-five millilitres and pH 7, before levelling off near pH 12.

### Periodic table excerpts and data tables

Do not describe a data table in alt text. Rebuild it as a real table with header cells. A screen reader can then navigate it cell by cell, which alt text can never allow.

### Assessment-safe alternative text

This is the problem most guidance ignores, and instructors are right to worry about it.

If a worksheet asks students to identify a functional group from a structure, alt text that names the functional group hands the answer to exactly the students the accommodation is meant to support.

**The principle: describe what a sighted student sees, not what they are asked to conclude.**

A sighted student looking at the diagram sees lines, atoms, and arrangement. They do not see a label reading “this is a ketone.” Your alt text should give the same raw information and no more.

| Task                          | Do not write                 | Write                                                                                     |
|-------------------------------|------------------------------|-------------------------------------------------------------------------------------------|
| Identify the functional group | “Structure showing a ketone” | “A five-carbon chain. Carbon three is double-bonded to an oxygen. All other carbons carry |

|                      |                                                |                                                                             |
|----------------------|------------------------------------------------|-----------------------------------------------------------------------------|
| Name the compound    | “Structure of 2-butanol”                       | hydrogens.”<br>“A four-carbon chain with a hydroxyl group on carbon two.”   |
| Balance the equation | “Balanced equation showing two moles of water” | “Hydrogen gas plus oxygen gas yields water.<br>Coefficients are not shown.” |
| Predict the product  | “Reaction producing ethanol”                   | “Ethene plus water over an acid catalyst yields an unlabelled product.”     |

Where the visual task cannot be preserved in text — genuine visual-discrimination items, for instance — the accessible route is an alternative assessment of the same underlying knowledge, not a compromised description. Ask the student to describe the structure in text and evaluate the description.

### Supplementary machine-readable notation

For chemistry specifically, consider providing SMILES or InChI strings alongside structures. They are compact, unambiguous, and let a student paste the structure into a molecular viewer or drawing tool.

Ethanol. SMILES: CCO

This supplements alternative text; it does not replace it. Most students will not know how to read SMILES, and it says nothing about how the structure was drawn.

### Length guidance

| Image type                          | Target                                      |
|-------------------------------------|---------------------------------------------|
| Decorative                          | Empty alt                                   |
| Simple molecule                     | One sentence                                |
| Reaction scheme                     | One to two sentences                        |
| Mechanism, apparatus, complex graph | Short alt plus a paragraph in the body text |
| Data table                          | Rebuild as a real table                     |

If your alt text exceeds roughly 150 characters, that is a signal to move the content into the body of the page.

### Checking your work

Read the alt text aloud without looking at the image, and ask whether you could reconstruct the structure from it. If you could not, a student cannot either.

Then run the accessibility checker in Word or your LMS. It will catch missing alt text. It cannot tell you whether the alt text is any good — only the read-aloud test does that.

## WCAG 2.1 criteria addressed

| Criterion                    | Level | Application                                                                     |
|------------------------------|-------|---------------------------------------------------------------------------------|
| 1.1.1 Non-text Content       | A     | Every informative image has a text alternative conveying equivalent information |
| 1.3.1 Info and Relationships | A     | Data presented visually as tables is marked up as tables                        |
| 1.4.1 Use of Color           | A     | Structural features are not distinguished by colour alone                       |
| 1.4.5 Images of Text         | AA    | Labels within diagrams are available as text                                    |

*Feedback on this guide is welcome and will be credited in future versions.*